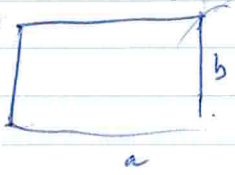
18.313 - Rota

Famous Unsolved Probs

1) rug



drop at random
n pennies

hard spheres prob

prob no 2 pennies shall overlap?

Solved for 4, not 5.

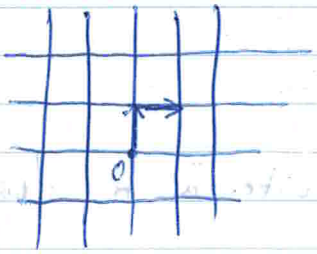
and, 1 dim needles $\equiv\}$ b

stick

ans $\dfrac{(a-nh)^n}{(a-h)^n}$

2) postage stamps, how many ways can you fold so only one stamp is visible

3) Random walk



take 4 sided fair die

↑ → ↓ ←

move according to direction die indicates

the prob: { non-self intersection random walk in n steps that
self-avoiding
n steps

p_n → that will not return to a vertex already traced.

$\lim_{n\to\infty} \sqrt[n]{p_n}$ exists

4) Cell Growth

take die of unexposed side and attach new squared to side indicated bumbup die

5) Percolation

trees in an orchard equally spaced

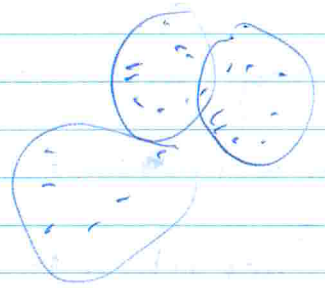
infected tree infects each of it neibs w/prob p and dies w/prob $q = 1-p$

What is critical prob.

6) Cluster Analysis

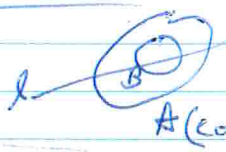
Collect Data

what precise criteria for clustering



Geometric Prob

Sylvesters thm



A (convex)

pt at random $\text{Prob}(p \in B \mid p \in A) = \dfrac{\text{area } B}{\text{area } A}$

$\text{prob}(\ell \cap B \neq \emptyset \mid \ell \cap A \neq \emptyset) = \dfrac{\text{perim } B}{\text{perim } A}$

PS #1

88

T.T.

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W11

P.28 #15 a) No the ~~probabilities~~ properties of being male and female are not independent.

b) Yes the property of being male and ~~female~~ married are independent

why? −2

P.28 #17 The events that he gets the scholarships are not independent. In effect, this means that should he get the first scholarship his chances of getting the 2nd are higher

$P(A) = 1/2$, $P(B) = 1/4$ $P(A \cap B) \neq P(A)P(B) \Rightarrow$ dep

P.62 #1

	H	T
H H H	123	
H H T	12	3
H T H	13	2
H T T	1	23
T H H	23	1
T T H	3	12
T H T	2	13
T H H	1	23

None of these satisfy the exclusion principles.

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+T.T

2 – Spelling

Essam Abadir

P.62 #2

	Box 1	2	3	4	5	6
1 1	2					
1 2	1	2				
1 3	1		2			
1 4	1			2		
1 5	1				2	
1 6	1					2
2 1	2	1				
2 2		2				
2 3		1	2			
2 4		1		2		
2 5		1			2	
2 6		1				2
3 1	2		1			
3 2		2	1			
3 3			2			
3 4			1	2		
3 5			1		2	
3 6			1			2
4 1	2			1		
4 2		2		1		
4 3			2	1		
4 4				2		
4 5				1	2	
4 6				1		2
5 1	2			1		
5 2		2		1		
5 3			2	1		
5 4				2		
5 5				1	2	
5 6				1		2
6 1	2				1	
6 2		2			1	
6 3			2		1	
6 4					2	1
6 5						2
6 6						1 2

P.62 #4

	Box #	Box #
AA	1, 2	
AB	1, 1	2, 1
AC	1, 1	3, 1
AD	1, 1	4, 1
AE	1, 1	5, 1
AF	1, 1	6, 1
BA	1, 1	2, 1
BB	2, 2	
BC	2, 1	3, 1
BD	2, 1	4, 1
BE	2, 1	5, 1
BF	2, 1	6, 1
CA	3, 1	1, 1
CB	3, 1	2, 1
CC	3, 2	
CD	3, 1	4, 1
CE	3, 1	5, 1
CF	3, 1	6, 1
DA	4, 1	1, 1
DB	4, 1	2, 1
DC	4, 1	3, 1
DD	4, 2	
DE	4, 1	5, 1
DF	4, 1	6, 1
EA	5, 1	1, 1
EB	5, 1	2, 1
EC	5, 1	3, 1
ED	5, 1	4,
EE	5, 2	
EF	5, 1	6,
FA	6, 1	1,
FB	6, 1	2,
FC	6, 1	3,
FD	6, 1	4,
FE	6, 1	5,
FF	6, 2	

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P.62 #51

$$\frac{8!}{5!\,2!\,1!} = \frac{8\cdot 7\cdot \cancel{6}^{3}}{\cancel{2}\cdot 1} = 168$$

8! ← games; 5! ← wins, 2! ← losses, 1! ← ties

P.63 #61

$$\binom{10}{4} = \frac{10\cdot 9\cdot 8\cdot 7}{4\cdot 3\cdot 2} = 210 + \binom{10}{5} + \binom{10}{6} = 672$$

10: # of people

4: # of spots undetermined after the 1st six are

5: # of spots if 5 in first and 5 in 2nd

6: # of spots if 4 in first 6 in second

P.63 #10 Let A be the event that no two students have the same birthdays. Let B be the event that no two students have adjacent birthdays.

Then we want $A \cup B$

$$P(A) = \frac{(365)_5}{(365)^5} \qquad P(B) = ?$$

The two events are disjoint so

$$P(A \cup B) = P(A) + P(B)$$

-10

-10

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P.64 #14

$$(.9)^4 + \left[(.9)^4(.1)\right]\binom{4}{1} = .92 = \text{prob he wins.}$$

assumption here is that Robin will never choose to miss unless he has hit his first four shots ∴ the term $(.9)^4$ represents that one situation, the term $(.9^4 \cdot .1)4$ represents the five diff. ways he can hit the bull's without hitting his first four shots.

P.68 #28

Prob of 1 six on 4 throws $= 1-(5/6)^4 = .5177469$

Prob of double six on 24 rolls $= 1-(35/36)^{24} = .4914039$

17.109 2/3/93

Hart's Response to the Rule-skeptic

1) Response to "Extreme position" (ie. there are no rules)

partic. to Holmes's prediction theory
a) If there are courts there are rules

2) More moderate version: there is only one rule
- defining a court.

Response - if rules + rules which count courts

re:resp - rules not recog. till courts apply them

re:re:resp - there there is a R.R.

3) No primary rules (Internal Pr of View).

7) Brownian Motion, or the Wiener process

take random walk use smooth curve (no jumps)

The Course

Unfair coin $\begin{cases} \text{heads} & p \\ \text{tails} & q \end{cases}$ Bernoulli Process

H 1
T 0

$(HHTHT\ldots) = (11010\ldots)$

$\omega \in \Omega$ = set of such sequences

sample pts

18.313 2/1/93

The Concept of Probability

Bernoulli Sample Space

biased coin: $\begin{cases} p \\ 1-p \end{cases}$

$w = \{H, T, T \ldots\} = \{1, 0, 0 \ldots\}$

$\{w\} = \Omega$ A sample space Ω is a set together with a family of subsets called events

satisfying:

1) Ω and $\emptyset$ are events
2) If A is an event $(\bar{A} = A^c)$ its complement.

$$A^c = \Omega \setminus A$$

is also an event

3) If A and B are events, then so are $A \cap B$ and $A \cup$
4) If $A_1, A_2, \ldots$ is a sequence of events, then $A_1 \cup A_2$ is also an event

---

A probability, P, on a sample space, Ω, is a function which assigns to each event a # btw 0 and 1.
subject to:

1) Prob. of $\emptyset$ is zero $P(\emptyset) = 0$
2) Prob of the complement of an event is 1 - prob of event $P(A^c) = 1 - P(A)$
3) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
4) If $A_1, A_2, \ldots$ is a sequence of events s.t. $A_i \cap A_j = \emptyset$ for $i \neq j$ then $P(A_1 \cup A_2 \cup \ldots) = P(A_1) + P(A_2) + \ldots$

Two events A and B (in a given sample space) are said to be <u>independent</u> when $P(A \cap B) = P(A)P(B)$

Three events A, B, C are indep when: $P(A \cap B \cap C) = P(A)P(B)P(C)$

$P(A \cap B) = P(A)P(B)$

$P(A \cap C) = P(A)P(C)$

etc.

similarly for more than 3 events

<u>Ex. 0</u>

"Picking a pt @ random in the unit square"

1st define events

2nd identify prob of events

1) find event whose prob is obv. by applying rules 1, 2, 3, 4.

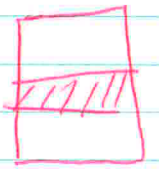
area strip is $l \times 1$ $\therefore$ $P(\text{strip base } l) = l$

l' (horiz. or vert).

$P(\text{rect. R of sides a and b}) = a \cdot b$

$P(R \cup C) = P(R) + P(C) - P(R \cap C)$

$(A \cap B) = R$ $\therefore$ $P(R) = P(A)P(B) = l \cdot l'$

The events C = and D =  are indep.

since $P_t = (x, y)$

C function of x and D function of y.

18.313

Ex. 1 · The Bernoulli Process

$$\Omega = \{ w = (1, 0, 1, 1, \ldots)$$

sample pts.

H_n = the event that the nth toss is Head.

= the set of all sequences (= sample pts) $w = (w_1, w_2, \ldots w_n)$

$w_i = \begin{cases} H \\ T \end{cases}$ for which $w_n = H$

$H_n^c = T_n$

* If k and n are diff. indicies then the events H_k and H_n should be independent.

* $P(H_n) = p$ hence $P(T_n) = q$

An event for the Bernoulli Process is an subset that can be obtained from the event H_n, $n = 1, 2, \ldots$ by (1)-(4) above

The Prob P for Bernoulli process is well defined for all events after setting:

1) $P(H_n) = p$, $n = 1, 2, \ldots$ and $P(H_k \cap H_n) = P(H_k) P(H_n)$, $k \neq n$
any set $\{H_{i_1}, H_{i_2}, \ldots, H_{i_j}\}$ is independent

If A and B are indep. events. then A and B^c are indep events.

$$\text{Prob}(A \cap B^c) = P(A - A \cap B) = P(A) - P(A \cap B) = P(A) - P(A)P(B)$$
$$= P(A)(1 - P(B)) = P(A)P(B^c).$$

Note: is a conseq. of the axioms we have:
if $C \subseteq A$ then $P(A - C) = P(A) - P(C)$
$P(C) + P(A - C) = P(A)$

Application: if Bern Process, events H_n, T_n are indep for $k \neq n$

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$P(\text{any intersection of H's and T's}) = p^{\text{no of H's}}\, q^{\text{no of T's}}$

def: a run, uninterrupted sequence of 1, or zero's

(could mean more than 2 H in a row, e.g. HHH)

A = event that a run of HH occurs before a run TT

$P(A) = ?$

Run

$p = \frac{1}{2}$ easy

$P(A) = \frac{1}{2}$ since $P(A) = P(A^c)$

18.313 2/3/43

Probability = Probability function

$P(\bigcup_i A_i) = \sum_i P(A_i)$ = $P(A_1) \cup P(A_2)$ ....

<u>Bernoulli Sample Space</u>

Ω = all tosses

Basic events: H_n = n-th toss results in H

if $h \neq n$ H_h and H_n are independent; $P(H_n) = p$ $P(H_k \cap H_n) = P(H_k)P$

$= p^2$

A = event that a run HH occurs before a run TT

$P(A) = ?$ $A = (H_1 \cap H_2) \cup (H_1 \cap T_2 \cap H_3 \cap H_4) \cup (T_1 \cap H_2 \cap H_3) \cup$

and or $(\ldots\ldots)$

More elegant Method: A

HH before a run TT

A run of HH before TT starting on nth toss

A_n = event that in $(w_n, w_{n+1}, \ldots$

$(w = (w_1, w_2, \ldots w_n, w_{n+1} \ldots)$

$\boxed{A = (A \cap H_1) \cup (A \cap T_1)}$ ☆

$A \cap H_1 = (H_1 \cap H_2) \cup (H_1 \cap A_2 \cap T_2)$; $A \cap T_1 = (T_1 \cap H_2 \cap H_3)$ ~~$\cup T_1 \cap H_2 \cap A$~~

$= T_1 \cap H_2 \cap A_2$

$P(A \cap T_1) = q P(H_2 \cap A_2)$

$P(A \cap H_1) = p^2 + p P(T_2 \cap A_2)$

Observe that $P(H_n \cap A_n) = P(H_1 \cap A)$; $P(T_n \cap A_n) = P(T_1 \cap A_1)$

~~$P(A \cap T_1) = qP$~~ $P(A \cap H_1) = p^2 + pq P(H_1 \cap A)$

$P(A \cap H_1) = p^2 / 1-pq$

$P(A \cap T_1) = \frac{qp^2}{1-pq}$

$P(A) = \frac{p^2 + qp^2}{1-pq}$

$$\underbrace{P(\text{neither } HH \text{ occurs before } TT \text{ nor } TT \text{ occurs before } HH)}_{B}$$

$P(B) = ?$ $\qquad B = H_1 \cap T_2 \cap H_3 \cap T_4 \cap \dots) \cup (T_1 \cap H_2 \cap T_3$

$$\lim_{n\to\infty} P(H_1 \cap T_2 \cap H_3 \cap \dots \cap T_{2n}) = \lim_{n\to\infty} p^n q^n = 0 \rightarrow P(B) = 0$$

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ex. <u>The uniform sample space</u>

pick a pt at random

elementary events $[a,b]$ and $P([a,b]) = b-a$ $\quad 0 \le a \le b \le 1$

A = event that the chosen # is rational i.e. $\frac{m}{n}$ where m, n, are integers.

$[a,a] = \{a\}$ $\quad P([a,a]) = 0$ $\quad \{r_1, r_2, \dots\}$ = set of all rational #'s

$$P(A) = P(r_1) + P(r_2) + \dots + \dots = 0 + 0 + \dots = 0 \qquad P(A) = 0$$

18.313 2/5/93

Finite Sample Spaces Ω = finite no. of sample spaces

Every subset is an event, and if $P(w)$ is given for every sample pt. $w \in \Omega$, then for every event A we have
$$P(A) = \sum_{w \in A} P(w)$$

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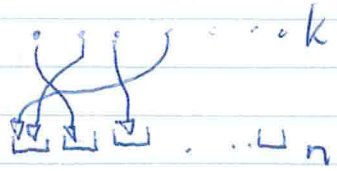
1) Maxwell-Boltzman statistics (sample space)

Balls and Boxes

Sample points



Balls

Urns

sample pt in this is a function from B to U

$\Omega = \{w = \text{function from } B \text{ to } U$

$|\Omega| = n^k$

$P(w) = \frac{1}{n^k}$; if boxes are labled $1, 2, \dots, n$ for every w we set $\theta_i(w)$ = no of balls in i_{th} box

$$\sum_{i=1}^{n} \theta_i(w) = k$$

$\theta_1, \dots, \theta_1$ = occupation nos.

# of one-to-one functions from B to the set $U = (n)_k = n \cdot (n-1)(n-2) \cdots (n$

$n_0 = 1$

Ex. A = event that no two out of k people have the same birthday.

$$P(A) = \frac{(365)_k}{(365)^k}$$

A = event that a k-letter word shall not repeat

How many balls in each Box?

event

$$P(\theta_1 = i_1, \theta_2 = i_2, \dots, \theta_n = i_n) = \frac{1}{n^k}\binom{k}{i_1, i_2, \dots, i_n}$$

multinomial coefficient

$r! = 1 \cdot 2 \cdot 3 \cdots r$

$= (r)_r$

$\Rightarrow \frac{1}{n^k} \cdot \frac{k!}{i_1! \, i_2! \cdots i_n!}$ if $i_1 + i_2 + \cdots + i_n = k$

0 otherwise (always take limiting case)

☆ $\boxed{0! = 1}$

Stoppers

. . . . . .

$\underbrace{\quad}_{i_1}$ $\underbrace{\quad}_{i_2}$

(Boxes sep by stoppers
n-1 stoppers since don't need first)

rearrange balls + let drop

hand-wave method

→ however, over counting

why? Because we don't care about order of balls in boxes. Hence we divide by $i_1, i_2, \dots$ to remove the overcounted combinations.

Sample Space of Dispositions ← way of plac balls in boxes where order counts of each box

. . . . . k B

eg

$\sqcup \sqcup \sqcup \sqcup \cdots$ n U

$|\Omega_D| = ? = n(n+1)(n+2)\cdots(n+k-1) = \langle n \rangle_k = n^k$

rising factorial

Ω_D

Proof by method of stoppers.
each ball added can be viewed as an extra stopper
each successive ball has one more option than the one

disposition
- function from
n U plus an
ordering of the balls
each box